Titanium Models A and B

Ground-truth comparison, probability review and future forecast analysis

This report compares available historical observations with predictions for both titanium models. It also presents the genuine future forecasts produced from the June 2026 source cutoff. Future target months do not yet have ground truth and therefore cannot be scored today.

Model	Prediction task	Outputs	How performance is judged
Model A	UK titanium import unit price	P05, P50 and P95 GBP/kg at H3, H6 and H12	P50 deviation, MAE/RMSE, direction accuracy, interval coverage and interval score
Model B	Supplier-share movement	Collapse, fall, stable or rise probabilities at H3, H6 and H12	Accuracy, balanced accuracy, macro F1, Brier score, log loss and calibration

Prepared from saved model evaluation outputs and future forecast files. Source data ends June 2026.

1. Data, training and evaluation periods

The two models use expanding-time evaluation, but their saved production pipelines do not use identical date blocks. The table below records the exact implementation rather than forcing one split definition onto both models.

Model	Stage	Dates used	Purpose
Model A	Development and blocked validation	Development through Jan 2022; blocked forecast origins Jan 2017-Jan 2022	Fit and select price algorithms without using future target values
Model A	Disruption test	Forecast origins Feb 2022-Dec 2023	Stress-test price forecasts across the disruption period
Model A	Post-disruption test	Forecast origins Jan 2024-Dec 2025	Test recent performance; H12 availability is limited by the Jun 2026 cutoff
Model B	Initial training	2010-2017, expanding forward	Fit the supplier-share movement classifier
Model B	Validation	Jan 2018-Jun 2021	Historical model selection and probability evaluation
Model B	Disruption test	Jul 2021-Jun 2023	Test class predictions under structural change
Model B	Post-disruption test	Jul 2023-latest completed target	Test recent generalisation

Historical comparison versus future prediction

Validation and test forecasts are generated as if the later observations were unknown, then compared with the values that subsequently occurred. This provides ground truth. Separately, both models contain forecasts from a June 2026 origin for September 2026, December 2026 and June 2027. Those future rows cannot yet be evaluated because the source dataset stops at June 2026.

The original statement 'training 2010-2017, then validation 2018-2021' accurately describes Model B. Model A's production_v6 saved evaluation uses the model-specific blocks shown above.

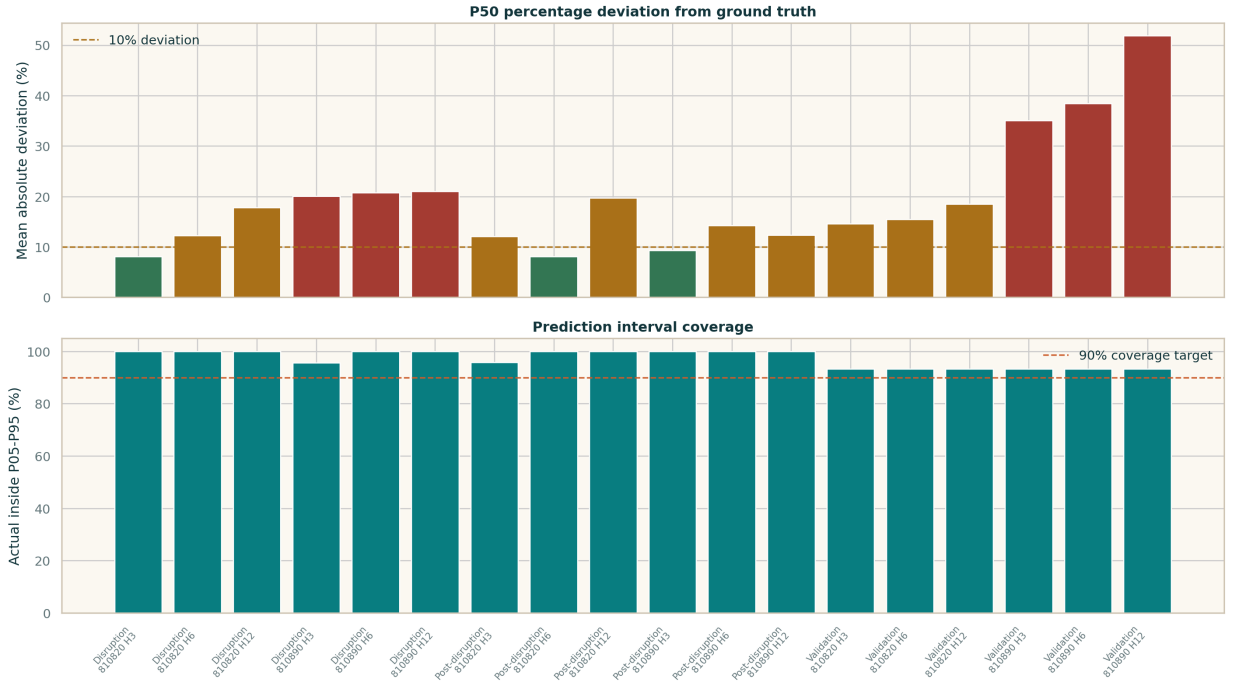
2. Model A price forecast performance

Model A predicts an import-price distribution for HS 810820 and HS 810890. P50 is the central prediction. Percentage deviation measures how far P50 is from the actual price. Coverage reports whether the actual price fell between P05 and P95.

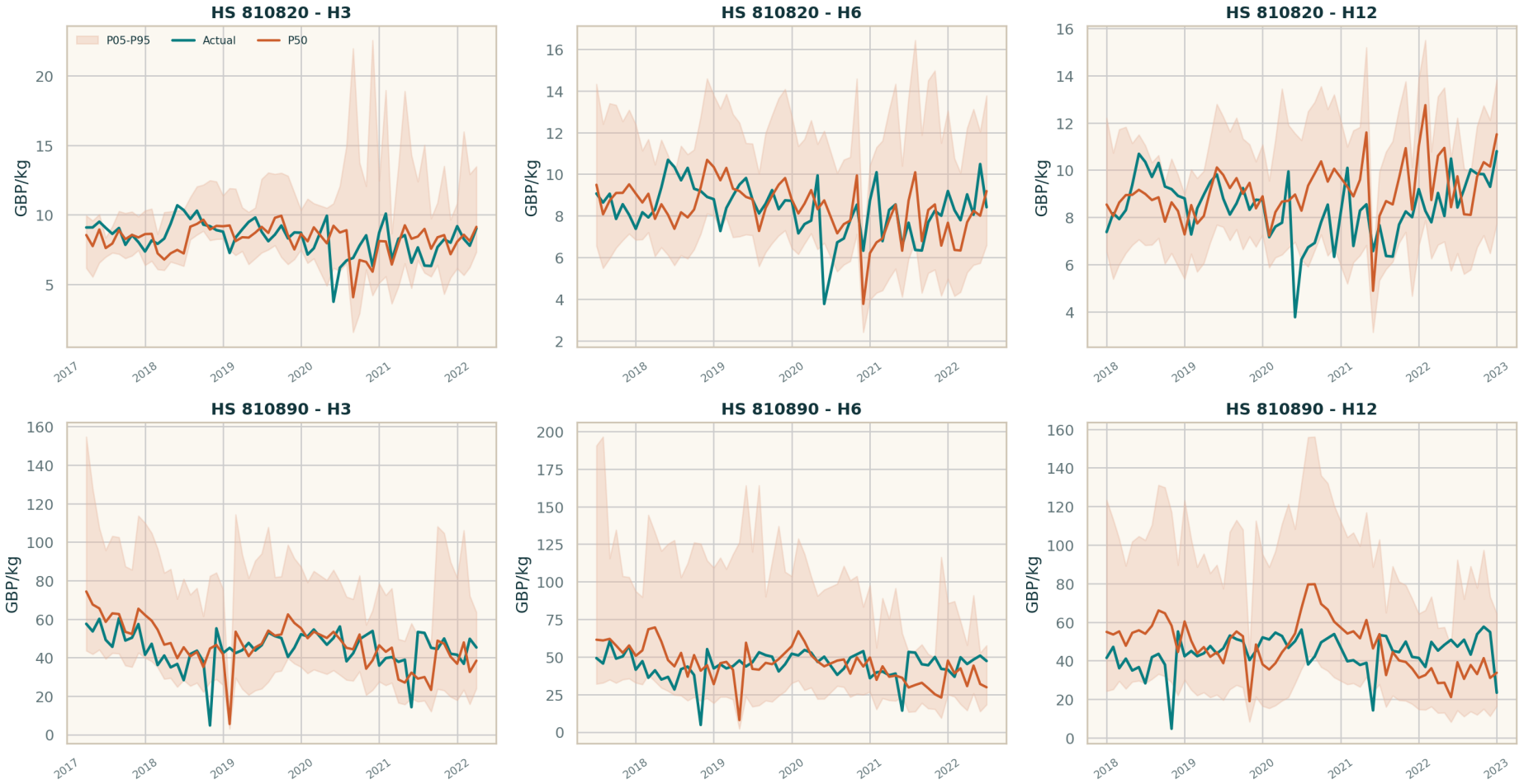
Period	Rows	Mean abs. deviation	Within 10%	Direction accuracy	P05-P95 coverage
Validation	360	29.0%	37.2%	48.6%	93.3%
Disruption test	138	16.7%	46.4%	50.0%	99.3%
Post-disruption test	132	12.3%	52.3%	50.8%	99.2%

Overall point accuracy improves in later periods: mean absolute deviation falls from 29.0% in validation to 16.7% during disruption and 12.3% post-disruption. The broad intervals cover 93.3%-99.3% of actual outcomes. This strong coverage is useful, but intervals that are very wide can be conservative rather than precise.

Model A error and uncertainty performance

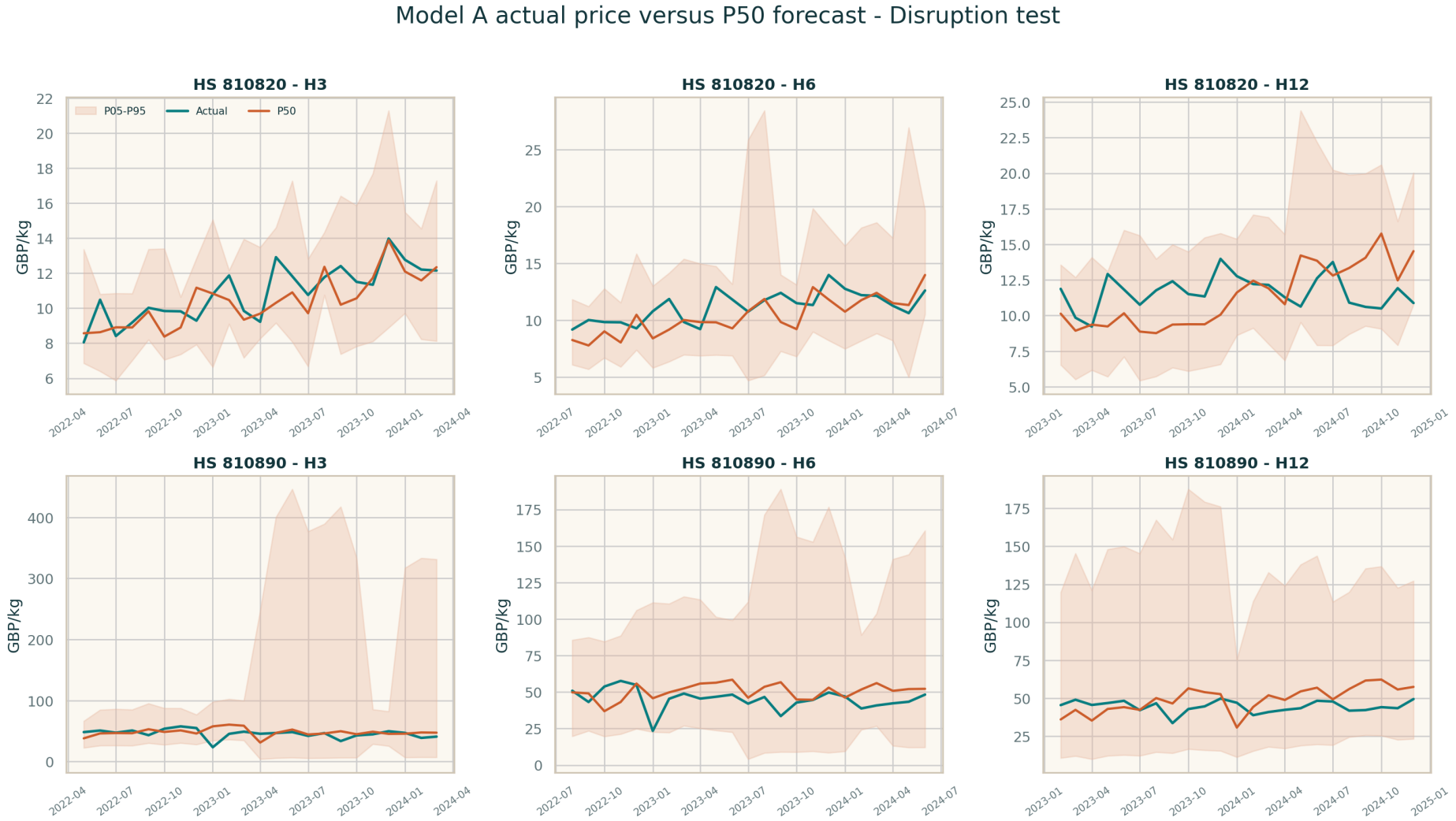


Model A validation: actual price versus P50



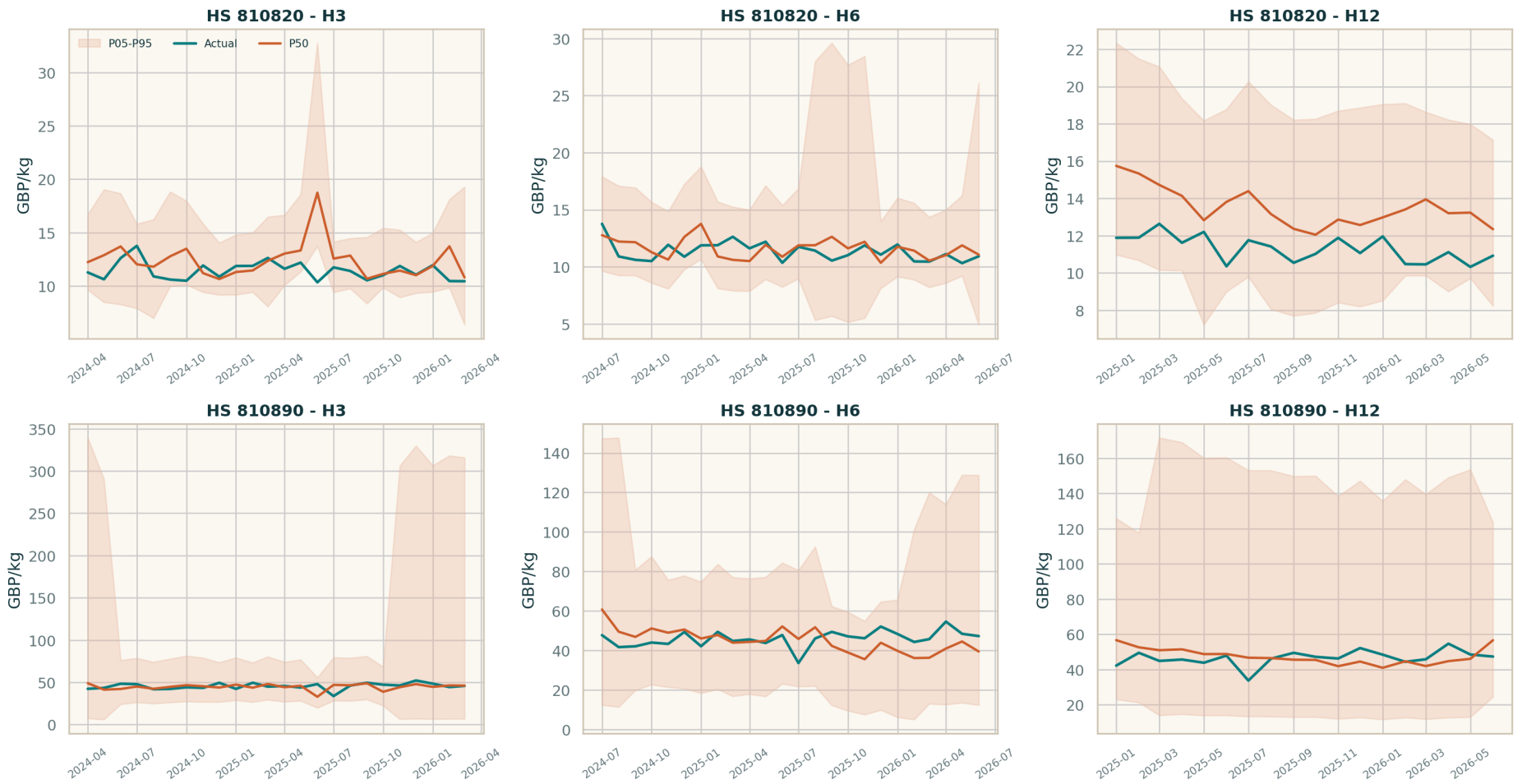
Each panel is one HS code and horizon. The shaded area is P05-P95; the teal line is actual price and the orange line is P50.

Model A disruption test: actual price versus P50



This period checks whether the model remains useful when market behaviour differs from the earlier development period.

Model A post-disruption test: actual price versus P50



This is the most recent historical evidence available for judging current price-forecast behaviour.

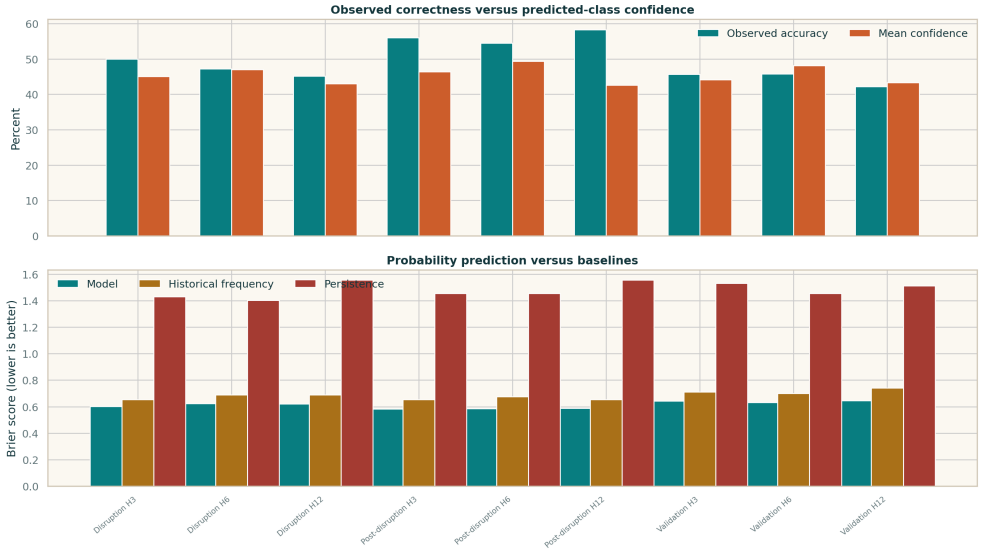
3. Model B supplier-shift performance

Model B classifies each supplier's future share movement as collapse, fall, stable or rise. The predicted-class confidence is the largest of the four class probabilities. It is not the same as observed accuracy and should not be presented as a guaranteed chance of correctness.

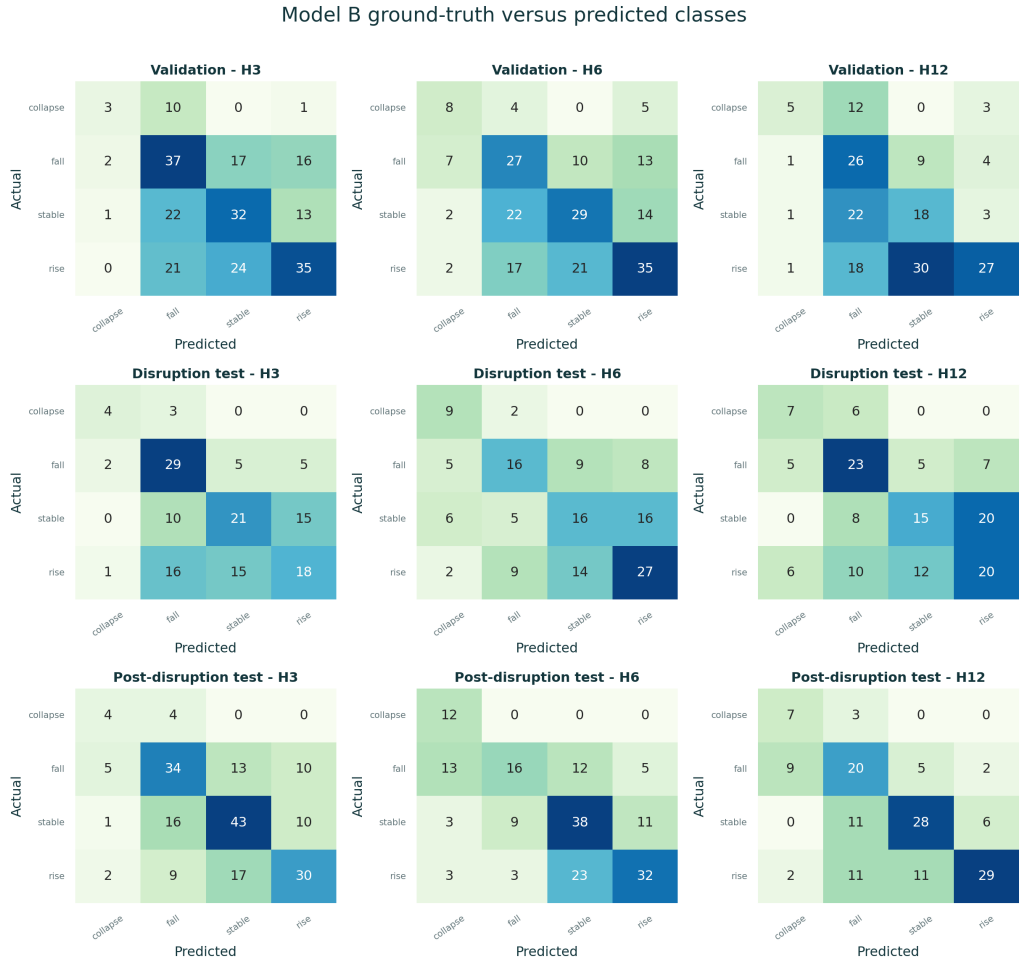
Period	Rows	Accuracy	Mean confidence	Balanced accuracy	Macro F1	Brier	ECE
Validation	630	44.8%	45.3%	42.9%	43.0%	0.640	0.028
Disruption test	432	47.5%	45.0%	50.9%	48.1%	0.616	0.036
Post-disruption test	522	56.1%	46.4%	58.9%	54.1%	0.585	0.054

Observed accuracy rises from 44.8% in validation to 47.5% in disruption and 56.1% post-disruption. The model's Brier score is lower than the saved historical-frequency and persistence baselines across the evaluated blocks. Calibration is still imperfect, especially in some horizon-specific reliability bins, so probability claims require caution.

Model B classification and probability performance

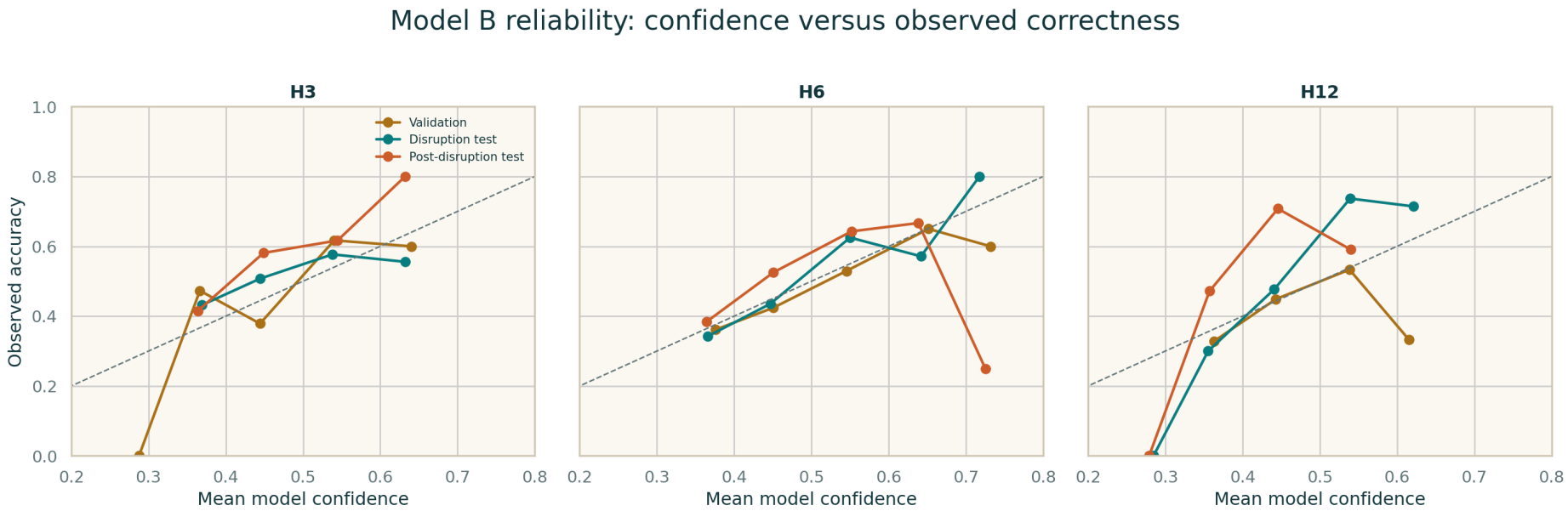


Model B ground truth versus predicted classes



Diagonal cells are correct classifications. Off-diagonal cells show which movement classes the model confuses.

Model B probability reliability

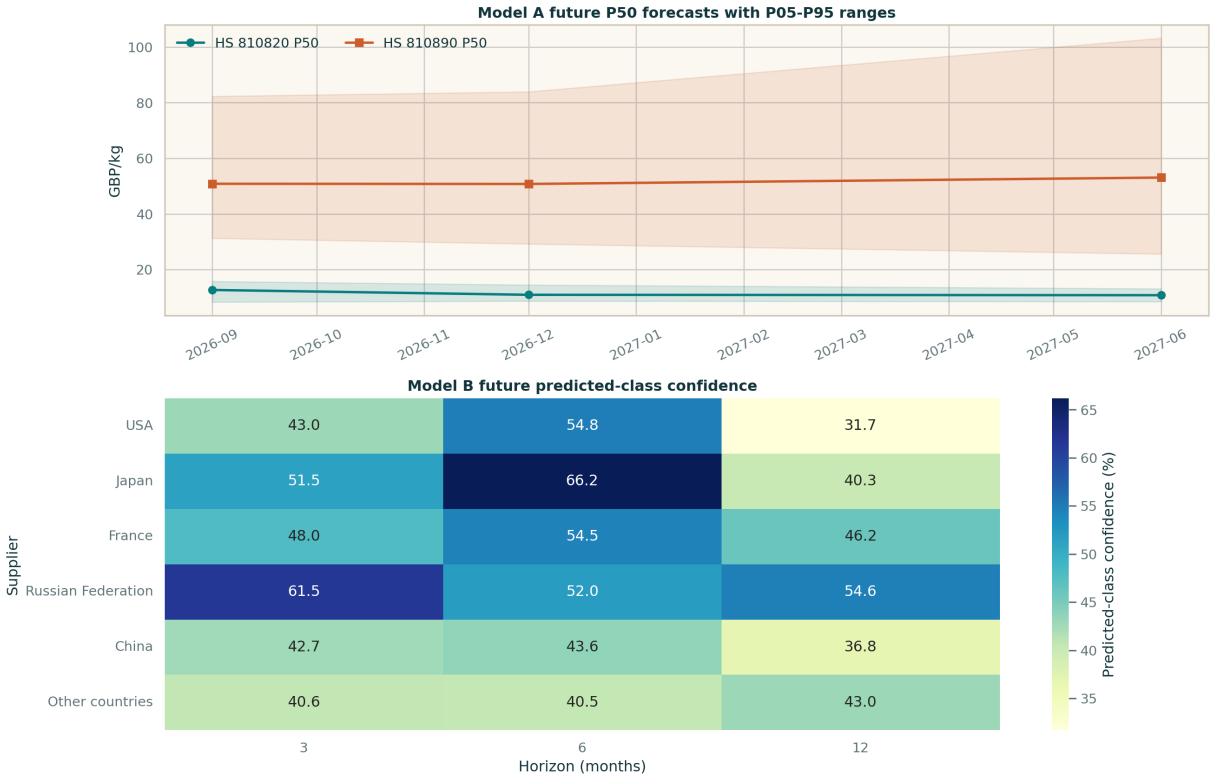


A perfectly calibrated model follows the dashed diagonal: predictions made with 60% confidence would be correct about 60% of the time. Visible departures from this line explain why calibration remains a limitation even when the classifier beats its probability baselines.

4. Future forecasts after the source cutoff

Yes, both saved pipelines generate later 2026 and 2027 forecasts. The current files use June 2026 as the forecast origin and produce H3, H6 and H12 targets: September 2026, December 2026 and June 2027. These rows are predictions only. They do not yet have actual values in the dataset and are therefore excluded from performance scoring.

Future forecasts from June 2026 origin



Model A future price forecasts

HS code	Target	Horizon	Model	P05	P50	P95	P50 change
810820	2026-09	H3	linear_quantile	GBP 8.30/kg	GBP 12.66/kg	GBP 15.77/kg	15.9%
810820	2026-12	H6	adaptive_no_change	GBP 8.66/kg	GBP 10.92/kg	GBP 14.48/kg	0.0%
810820	2027-06	H12	random_forest_distribution	GBP 8.49/kg	GBP 10.78/kg	GBP 13.11/kg	-1.3%
810890	2026-09	H3	elastic_net_residual	GBP 31.26/kg	GBP 50.83/kg	GBP 82.28/kg	7.5%
810890	2026-12	H6	linear_quantile	GBP 29.22/kg	GBP 50.76/kg	GBP 83.93/kg	7.3%
810890	2027-06	H12	elastic_net_residual	GBP 25.60/kg	GBP 53.06/kg	GBP 103.19/kg	12.2%

The P05-P95 width grows materially for HS 810890, reflecting substantial uncertainty. P50 is not required to rise with the forecast horizon, and interval bounds are not required to be monotonic. They describe different horizon-specific forecast distributions.

Model B future supplier-shift probabilities

Supplier	Target	Horizon	Predicted class	Confidence	Collapse	Fall	Stable	Rise
China	2026-09	H3	fall	42.7%	0.6%	42.7%	28.7%	28.0%
France	2026-09	H3	stable	48.0%	0.3%	31.4%	48.0%	20.3%
Japan	2026-09	H3	rise	51.5%	4.6%	26.8%	17.2%	51.5%
Other countries	2026-09	H3	fall	40.6%	3.0%	40.6%	23.0%	33.3%
Russian Federation	2026-09	H3	stable	61.5%	0.3%	16.2%	61.5%	22.0%
USA	2026-09	H3	fall	43.0%	25.6%	43.0%	14.2%	17.2%
China	2026-12	H6	fall	43.6%	1.0%	43.6%	34.4%	21.0%
France	2026-12	H6	stable	54.5%	0.6%	24.6%	54.5%	20.4%
Japan	2026-12	H6	rise	66.2%	4.8%	18.6%	10.4%	66.2%
Other countries	2026-12	H6	fall	40.5%	11.6%	40.5%	21.0%	26.8%
Russian Federation	2026-12	H6	stable	52.0%	0.5%	21.3%	52.0%	26.3%
USA	2026-12	H6	collapse	54.8%	54.8%	19.4%	11.8%	14.0%
China	2027-06	H12	fall	36.8%	4.7%	36.8%	32.7%	25.8%
France	2027-06	H12	stable	46.2%	1.5%	26.3%	46.2%	26.1%

Supplier	Target	Horizon	Predicted class	Confidence	Collapse	Fall	Stable	Rise
Japan	2027-06	H12	rise	40.3%	6.8%	27.6%	25.4%	40.3%
Other countries	2027-06	H12	fall	43.0%	5.1%	43.0%	28.5%	23.4%
Russian Federation	2027-06	H12	stable	54.6%	1.2%	16.7%	54.6%	27.4%
USA	2027-06	H12	collapse	31.7%	31.7%	28.0%	13.8%	26.6%

5. Recent completed ground-truth examples

Model A: latest available completed comparisons

Split	HS	H	Origin	Target	Actual	P50	Abs. deviation	Covered
Post-disruption test	810820	H6	2025-12	2026-06	GBP 10.92/kg	GBP 11.06/kg	1.3%	Yes
Post-disruption test	810820	H12	2025-06	2026-06	GBP 10.92/kg	GBP 12.34/kg	13.0%	Yes
Post-disruption test	810890	H6	2025-12	2026-06	GBP 47.31/kg	GBP 39.49/kg	16.5%	Yes
Post-disruption test	810890	H12	2025-06	2026-06	GBP 47.31/kg	GBP 56.60/kg	19.6%	Yes
Post-disruption test	810820	H6	2025-11	2026-05	GBP 10.32/kg	GBP 11.88/kg	15.1%	Yes
Post-disruption test	810820	H12	2025-05	2026-05	GBP 10.32/kg	GBP 13.23/kg	28.2%	Yes
Post-disruption test	810890	H6	2025-11	2026-05	GBP 48.47/kg	GBP 44.57/kg	8.1%	Yes
Post-disruption test	810890	H12	2025-05	2026-05	GBP 48.47/kg	GBP 46.01/kg	5.1%	Yes
Post-disruption test	810820	H6	2025-10	2026-04	GBP 11.11/kg	GBP 11.02/kg	0.8%	Yes
Post-disruption test	810820	H12	2025-04	2026-04	GBP 11.11/kg	GBP 13.20/kg	18.8%	Yes
Post-disruption test	810890	H6	2025-10	2026-04	GBP 54.64/kg	GBP 41.00/kg	25.0%	Yes
Post-disruption test	810890	H12	2025-04	2026-04	GBP 54.64/kg	GBP 44.71/kg	18.2%	Yes

Model B: latest available completed comparisons

Split	Supplier	H	Origin	Target	Actual	Predicted	Correct	Confidence
Post-disruption test	China	H3	2026-03	2026-06	fall	fall	Yes	60.0%
Post-disruption test	China	H6	2025-12	2026-06	rise	stable	No	61.5%
Post-disruption test	China	H12	2025-06	2026-06	rise	stable	No	34.5%
Post-disruption test	France	H3	2026-03	2026-06	stable	stable	Yes	57.6%
Post-disruption test	France	H6	2025-12	2026-06	rise	stable	No	66.7%
Post-disruption test	France	H12	2025-06	2026-06	stable	stable	Yes	44.5%
Post-disruption test	Japan	H3	2026-03	2026-06	fall	rise	No	37.9%
Post-disruption test	Japan	H6	2025-12	2026-06	collapse	collapse	Yes	53.8%

Split	Supplier	H	Origin	Target	Actual	Predicted	Correct	Confidence
Post-disruption test	Japan	H12	2025-06	2026-06	fall	fall	Yes	34.5%
Post-disruption test	Other countries	H3	2026-03	2026-06	stable	fall	No	44.9%
Post-disruption test	Other countries	H6	2025-12	2026-06	rise	rise	Yes	56.2%
Post-disruption test	Other countries	H12	2025-06	2026-06	fall	fall	Yes	34.1%

The Excel workbook contains all 630 scored Model A rows and all 1,584 Model B rows, not only these recent examples. Use the detail sheets to filter by evaluation period, HS code, supplier, origin month, target month and horizon.

6. Conclusions and limitations

Question	Evidence-based conclusion
Can the models be compared with ground truth?	Yes. Historical validation, disruption and post-disruption rows include actual outcomes and predictions.
Is there a percentage deviation for Model A?	Yes. Signed and absolute deviation percentages are calculated per row. The workbook also reports within-5%, within-10% and within-20% rates.
Is Model B's probability the chance it is correct?	It is model confidence, not a guaranteed chance. Observed accuracy and calibration plots must be read beside it.
Are future 2026/2027 forecasts evaluated?	No. The future target months occur after the June 2026 source cutoff, so ground truth is not available yet.
What is strongest in Model A?	Post-disruption point deviation is lower than validation and P05-P95 coverage remains high.
What is strongest in Model B?	Post-disruption accuracy and balanced accuracy improve, and Brier score beats the saved baselines.
What remains a limitation?	Model A intervals can be wide; Model B probability calibration is uneven and future forecasts require later outcome monitoring.

Demo interpretation: Model A is suitable for demonstrating probabilistic price forecasts with explicit uncertainty. Model B is suitable for research/demo use when the calibration warning remains visible. Neither model's unobserved future outputs should be described as validated outcomes until actual data arrives.

Files created with this report

Titanium_Model_A_B_Result_Analysis.xlsx contains the summary, all ground-truth rows, future forecasts and metric definitions. The RESULT ANALYSIS folder also contains the analysis CSV files and standalone PNG visualizations used in this report.